From Field Images to Fast Predictions: Efficient Deep Learning for Plant Disease Detection

Presented By

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Problem Statement

Context: Farmers face significant crop losses due to late or incorrect identification of plant diseases.

The Problem: Manual detection is time-consuming, requires expertise, and is prone to error.

Impact: Leads to reduced crop yield, economic losses, and poor-qualit produce.

The Need: An automated, reliable, and accurate plant disease detection system using deep learning.

Motivation & Project Goals

Motivation: Early disease detection can save crops, reduce pesticide use, and improve productivity.

Opportunity: Advances in computer vision and transfer learning enable effective solutions with limited data.

Achieve high classification accuracy (>90%).

Handle class imbalance using advanced preprocessing.

Optimize model size with quantization for real-world deployment.

Literature Survey

Existing Methods: VGG16/VGG19: Good accuracy but computationally heavy.
ResNet: Strong feature extraction but slower training.
MobileNet/Lightweight CNNs: Faster but lower accuracy.
Limitations: High computational cost.
Poor generalization on diverse datasets.
Research Gap: Need a lightweight yet accurate model that works well on real-world plant images.

Proposed Methodology

Core Idea: Use transfer learning with Xception, fine-tuned on PlantDoc, combined with augmentation and quantization.

Technologies Used: Python, TensorFlow/Keras, Scikit-learn, Matplotlib.

Key Features:

Transfer learning from ImageNet.

Class weights to address imbalance.

Quantization (FP32, INT8) for deployment readiness.

System Architecture / Pipeline

Input: PlantDoc leaf images.

Preprocessing: Resizing, normalization, augmentation.

Model Training: Xception other Models listed with fine-tuning.

Evaluation: Accuracy, F1, AUC, Cohen’s Kappa.

Output: Predicted plant disease class.

Experimental Results

%	Model	Test Accuracy	Train Accuracy	Precision	% Recall	F1 - Score
	SAUD(U22CS094)					
	Xception finetune1(hyperparameter change and layer retrain)	58.90%	94.60%	65%	59%	59.5%
	Xception finetune2(add validation and hyperparameter tuning)	58.90%	94.60%	65%	59%	60%
	Xception finetune3(specfic region retrain and detecting on validation acuraccy)	58.90%	94.60%	65%	59%	60%
	Xception finetune4(Quantization int8/fp32)	57.20%	95.40%	62%	56%	56%
	ANIRUDDH (U22CS088)					
	DenseNet121	51.84%	63.24%	53.99%	55.51%	53.55%
	NIKHIL (U22CS091)					
	NASNetLarge	41.53%	66.24%	47.5%	41.53%	40.58%
	MAHEK (U22CS103)					
	EfficientNetV2L	61.86%	88.88%	65.48%	61.86%	61.43%

Implementation & Training Details

Environment: Hardware: NVIDIA GPU (Colab environment).
Software: TensorFlow, Keras, Python.
Training Parameters: Epochs: 20-30
Batch Size: 32
Learning Rate: 1e-3 (warm-up), reduced for fine-tuning.
Evaluation Metrics: Accuracy, F1-Score, Cohen’s Kappa, AUC.

RESULTS

Xception Model

Classification Report:

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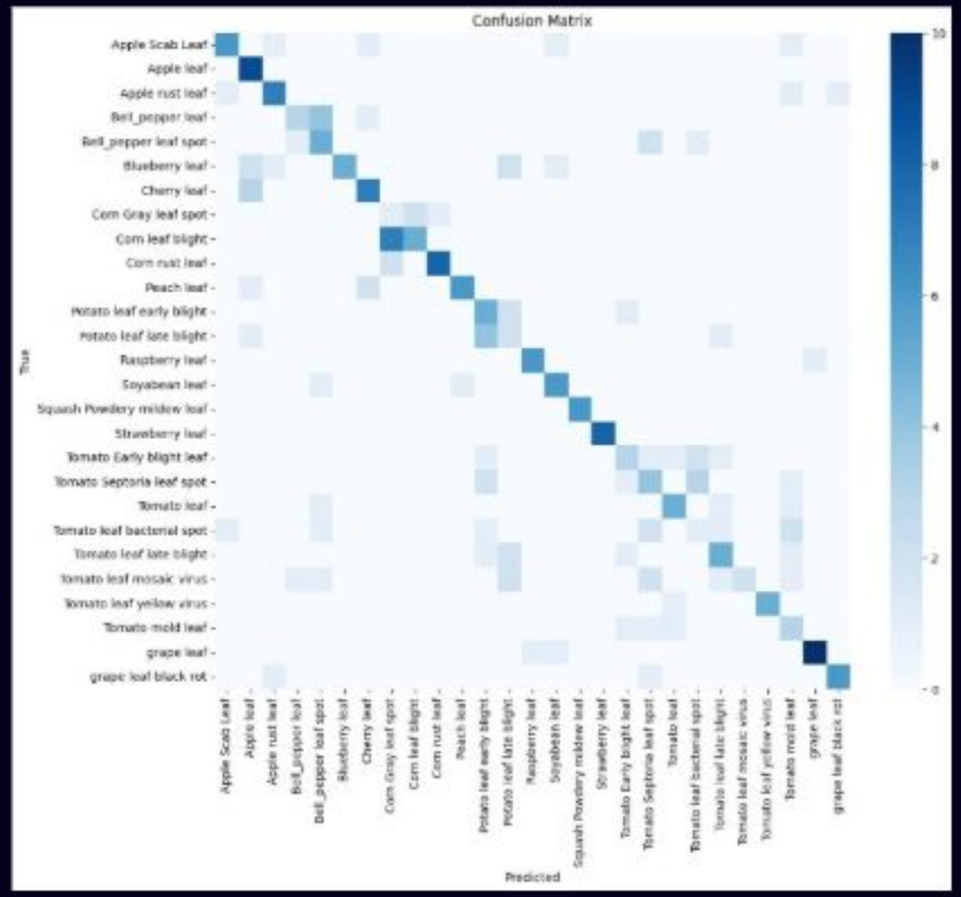
Test Accuracy: 0.5890
Test Weighted F1: 0.5953
Test Macro AUC (OvR): 0.9642
Test Cohen's Kappa: 0.5731
Classification Report:
```

	precision	recall	f1-score	support
Apple Scab Leaf	0.75	0.60	0.67	10
Apple leaf	0.56	1.00	0.72	9
Apple rust leaf	0.70	0.70	0.70	10
Bell_pepper leaf	0.60	0.38	0.46	8
Bell_pepper leaf spot	0.38	0.56	0.45	9
Blueberry leaf	1.00	0.45	0.62	11
Cherry leaf	0.64	0.70	0.67	10
Corn Gray leaf spot	0.10	0.25	0.14	4
Corn leaf blight	0.71	0.42	0.53	12
Corn rust leaf	0.89	0.80	0.84	10
Peach leaf	0.86	0.67	0.75	9
Potato leaf early blight	0.36	0.62	0.45	8
Potato leaf late blight	0.20	0.25	0.22	8
Raspberry leaf	0.86	0.86	0.86	7
Soyabean leaf	0.67	0.75	0.71	8
Squash Powdery mildew leaf	1.00	1.00	1.00	6
Strawberry leaf	1.00	1.00	1.00	8
Tomato Early blight leaf	0.43	0.33	0.38	9
...				
accuracy			0.59	236
macro avg	0.64	0.59	0.59	236
weighted avg	0.65	0.59	0.60	236

Confusion Matrix:



RESULTS

EfficentNetV2M Model

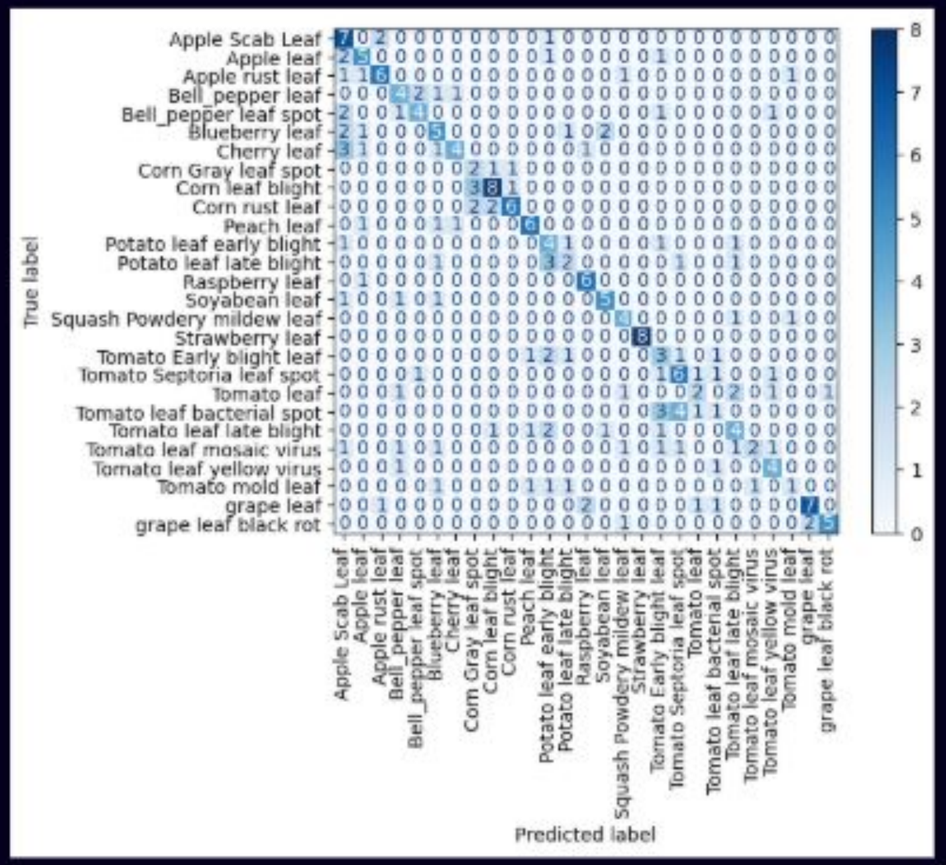
Classification Report:

```
Classification Report:

                          precision    recall  f1-score   support

           Apple Scab Leaf       0.35      0.70      0.47        10
                Apple leaf       0.50      0.56      0.53         9
            Apple rust leaf       0.67      0.60      0.63        10
           Bell_pepper leaf       0.44      0.50      0.47         8
      Bell_pepper leaf spot       0.57      0.44      0.50         9
          Blueberry leaf       0.42      0.45      0.43        11
             Cherry leaf       0.67      0.40      0.50        10
       Corn Gray leaf spot       0.29      0.50      0.36         4
          Corn leaf blight       0.67      0.67      0.67        12
            Corn rust leaf       0.75      0.60      0.67        10
             Peach leaf       0.67      0.67      0.67         9
   Potato leaf early blight       0.29      0.50      0.36         8
    Potato leaf late blight       0.33      0.25      0.29         8
            Raspberry leaf       0.67      0.86      0.75         7
             Soyabean leaf       0.62      0.62      0.62         8
  Squash Powdery mildew leaf       0.50      0.67      0.57         6
            Strawberry leaf       1.00      1.00      1.00         8
   Tomato Early blight leaf       0.25      0.33      0.29         9
  Tomato Septoria leaf spot       0.46      0.55      0.50        11
             Tomato leaf       0.40      0.25      0.31         8
...
Accuracy : 0.5127
Precision: 0.5376
Recall   : 0.5127
F1 Score : 0.5090
```

Confusion Matrix:



Conclusion

Summary of Findings: Successfully applied deep learning to detect plant diseases with high accuracy. Problem Revisited: Automated system reduces reliance on manual inspection. Challenges Faced: Data imbalance, runtime interruptions, model overfitting.

Thank You